

Coverage Is Not Predictive Fidelity: A Split-Landscape Analysis of Data Condensation for Gradient-Boosted Trees

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Abstract—

Data condensation replaces a large training set with a small surrogate so that a model retrained on it preserves downstream predictive behaviour. For gradient-boosted decision trees (GBDTs), one of the strongest practical model classes for tabular problems, many coreset and distillation methods now exist, but they are compared mostly by retrained AUROC or related task metrics. These scores are silent about mechanism: they do not reveal whether a condensed set preserves the split structure the booster would have learned or merely covers the input space, and they offer no way to screen candidates without retraining on each one.

We take a measurement-first view. A gradient-boosted tree commits to its tree structure through its split-selection landscape, while predictive behaviour also depends on leaf values and boosting dynamics. We formalize a no-candidate-retraining comparison of full and condensed split-gain landscapes as the Split-Landscape Discrepancy (SLD) and use it to study a broad slate of condensation methods (classical coresets, gradient- and gain-based selectors, and distribution-matching distillation) under a common protocol across many tabular datasets.

The analysis produces three principal results. In the small-to-mid-size regime where condenser choice changes AUROC, method-level split-regret is strongly rank-correlated with downstream AUROC (Spearman $\rho = -0.87$), making SLD useful for screening out bad surrogates and ranking method families, but not for single-dataset top-1 selection; at very large scale, random sampling becomes difficult to beat and root-regret loses fine ranking resolution as reasonable surrogates clear the dominant split margins. Decomposing error into structural and leaf-estimate components explains a counterintuitive failure: a method that greedily maximizes a specific high-gain split-threshold coverage objective is among the weakest performers, because unweighted coverage and predictive fidelity are different objectives. Finally, learned tree structure shows a margin-linked degradation pattern under landscape perturbation. We also delimit where the lens does not extend, including regression tasks and hyperparameter transfer, and release the benchmark and diagnostic as reusable tools.

Index Terms—data condensation; gradient-boosted decision trees; coreset selection; tabular data; no-candidate-retraining diagnostics

I. INTRODUCTION

ML engineers and data scientists who train gradient-boosted models on tabular data face a recurrent resource

tension. A single retraining run on a dataset of hundreds of thousands of rows can consume substantial compute; in our scale audit, condensed retraining reaches $160\times$ – $5,734\times$ speedups at the largest training sizes. Repeated runs for hyperparameter search, model versioning, or data-sharing workflows multiply that cost many times over. Condensation is not itself a privacy or compliance guarantee, but a small surrogate can be useful when combined with the appropriate governance or privacy mechanisms. The natural remedy, replacing the full dataset with a small, high-fidelity surrogate, is already practiced informally through random subsampling and stratified sampling. What is missing is a principled way to answer the question that matters most *before* committing to retraining on a surrogate: does this condensed set faithfully preserve the structure that the booster would have learned from the original data?

The dataset condensation and coreset-selection literature has produced a rich supply of methods to construct small surrogates [1]–[4]. For tabular data specifically, several methods have recently been adapted or purpose-built for non-image settings [5], [6]. Yet the evaluation paradigm is almost universally the same: train a fresh model on the condensed set, measure downstream predictive performance, and compare. This protocol has two practical shortcomings. First, it requires precisely the retraining one hoped to avoid. Second, it collapses a mechanistic question, namely *why* a condensed set works or fails, into a scalar outcome that gives no guidance for debugging or selection.

Gradient-boosted decision trees [7]–[9] offer a distinctive opportunity to break this cycle. A GBDT tree structure is determined by split-selection decisions: at each node, the algorithm chooses the feature and threshold that maximize the regularized gain criterion (Section III). Predictive behaviour also depends on leaf values, learning rate, boosting rounds, missing-value handling, and regularization. The gain criterion is histogram-sufficient once first- and second-order loss statistics are fixed; those statistics may come from a root base score or from a reference booster. Comparing the histogram landscapes of the full and condensed datasets is therefore a no-candidate-retraining screening diagnostic, not a claim that no model state is ever used.

We introduce the *Split-Landscape Discrepancy* (SLD),

which formalizes this comparison as an ℓ_p distance between the split-gain landscapes of a full dataset and its surrogate (Section III). Using SLD as a measurement instrument, we conduct a systematic comparative evaluation of 17 condensation methods, spanning classical coresets, gradient-based selectors, distribution-matching distillation, and gain-path distillation, under a unified protocol across a broad collection of tabular binary classification datasets. The analysis is a measurement study; we do not propose a new condensation method and we do not claim a new state of the art.

Our analysis surfaces three empirical findings of practical consequence. At the method-mean level, split-regret strongly ranks method families by downstream AUROC in the small-to-mid-size benchmark, enabling screening before retraining on every candidate. A provably coverage-optimal selector is simultaneously among the weakest methods by predictive performance, a demonstration that coverage and structural fidelity are independent objectives. At deployment scale, random sampling becomes difficult to beat, and SLD is better read as a rejection signal than as a single-best selector.

Contributions.

- **The SLD diagnostic.** We introduce the Split-Landscape Discrepancy, a no-candidate-retraining metric that quantifies how faithfully a condensed dataset reproduces the GBDT split-gain landscape of the full data under a fixed prediction state (Section III).
- **Systematic comparative analysis.** We present a 17-method evaluation specifically for GBDT-on-tabular settings, pooling results over thousands of dataset $\times$ budget $\times$ seed configurations and measuring both downstream predictive performance and landscape fidelity (Sections IV and V).
- **Three empirically grounded insights.** A single landscape score is method-level rank-correlated with AUROC at Spearman $\rho = -0.87$ in the small-to-mid-size benchmark; within-dataset trends are generally negative after controlling for dataset scale. Coverage and predictive fidelity are decoupled: a $(1-1/e)$ -optimal coverage selector is among the weakest downstream performers, explained by a reference-chain decomposition into structure and leaf-estimate gaps. Structure is robust to perturbation below a margin-linked threshold; the dose is a fraction of the split margin, not a fraction of intentionally corrupted splits (Section V).
- **Scope boundaries.** We document where the lens does not extend: regression tasks (structure-dominated failure), hyperparameter-rank transfer (near-zero Jaccard), and no-retraining selection of the single best method (top-3 rate 0.44), measured under the same protocol (Section VI).
- **Released artifacts.** The benchmark protocol, SLD implementation, processed tables, seeds, command lines, and plotting scripts will be released in a sep-

arate single-blind artifact repository: <https://github.com/anonymous-icdm2026/sld-gbdt-condensation>.

II. RELATED WORK

a) Dataset distillation and condensation.: Dataset distillation [1] and its successors optimise a small synthetic set so that a model retrained on it matches one trained on the full data, using gradient matching [2], distribution matching [10], or trajectory matching [11]. Comprehensive surveys [12]–[14] emphasize downstream evaluation, mostly on image benchmarks, and do not provide a GBDT-specific split-landscape fidelity metric. Extending distillation to tabular data has attracted recent attention [5], [15]–[17], but these works retain the retrain-and-evaluate paradigm. Our contribution is orthogonal: rather than a new condensation method, we introduce a no-candidate-retraining diagnostic that measures whether *any* condensed set, regardless of the method that produced it, preserves the GBDT split-gain landscape.

b) Coreset selection.: Classical coreset methods select real training examples by geometric or gradient-based criteria: herding [3] matches distribution moments; k-center [18] minimises the maximum coverage distance; CRAIG [4] and GRAD-MATCH [19] match gradient norms via submodular maximisation; EL2N and GraNd [20] use early-training gradient norms. All of these methods were designed for and evaluated on differentiable models. Translating their selection signals to the GBDT setting is non-trivial because GBDTs are not differentiable with respect to the training set; we include adapted variants in our 17-method benchmark and show that their landscape fidelity, measured by SLD, predicts their tabular AUROC. The GBDT-native subsampling methods GOSS [9] and MVS [21] operate per boosting round rather than producing a static condensed set; MVS is the theoretically closest prior work, as it explicitly minimises the variance of the split-scoring estimator. Data valuation [22], [23] and influence-function methods offer related but complementary perspectives on data contribution; they often require retraining on many subsets or differentiable-model assumptions and are not directly applicable in their standard form to histogram GBDTs.

c) Why GBDTs for tabular data.: Gradient-boosted decision trees [7]–[9], [24] remain one of the strongest practical model classes for tabular data [25], [26]. This motivates a condensation evaluation framework specific to tree-structured inductive bias. The split-gain criterion [27], [28] aggregates per-bin gradient statistics, and the resulting landscape fully determines which feature and threshold a node selects, a structure that has no direct analogue in differentiable models. Our SLD exploits this structure to build a measurement instrument that is GBDT-mechanism-specific: it explains *why* a condensed set preserves tree behaviour, not merely whether it does.

d) The C^2TC contrast.: The most closely related recent work is C^2TC [6] (ICDE 2026), a training-free

tabular condensation *method*: it optimises class allocation and feature representation via class-adaptive clustering to produce a condensed set efficiently. Our work is fundamentally different in intent. C²TC is a *condenser*: it produces a surrogate dataset. SLD is a *measurement instrument*: it audits the fidelity of any surrogate, including one produced by C²TC, with respect to the GBDT split-gain landscape. Where C²TC is model-agnostic in its clustering objective, SLD is GBDT-mechanism-specific: it reads the landscape from histogram statistics under a fixed prediction state without retraining on each candidate. We do not benchmark C²TC as a baseline in the current grid because the experiment package was frozen around methods for which we had compatible, reproducible implementations. The artifact repository exposes the same condenser interface for adding C²TC-generated sets; SLD can then serve as an evaluation layer or as a selection criterion among its class-allocation configurations.

III. BACKGROUND AND THE SLD DIAGNOSTIC

A. GBDT Split-Gain Landscape

Let the full training dataset be $D = \{(x_i, y_i)\}_{i=1}^N$ with features $x_i \in \mathbb{R}^F$ and labels y_i . A gradient-boosted decision tree (GBDT) fits an additive ensemble of trees by performing, at each node, a greedy split that maximises the regularised gain [7], [8]. Given first- and second-order loss statistics $g_i = \partial_{\hat{y}} \ell(\hat{y}_i, y_i)$ and $h_i = \partial_{\hat{y}}^2 \ell(\hat{y}_i, y_i)$ for each instance i , the gain of partitioning the instance set I at a node into left and right children I_L, I_R is

$$\mathcal{G}(I_L, I_R) = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma, \quad (1)$$

where $\lambda \geq 0$ is the ℓ_2 leaf-weight regulariser and $\gamma \geq 0$ is the minimum gain threshold [8]. Modern histogram-based GBDTs [9], [24] discretise each feature f into B bins; per-bin sums of (g, h) suffice to evaluate every candidate split.

Lemma 1 (Histogram sufficiency; standard, e.g. [9], [21]). *Given the current gradients and Hessians, a histogram GBDT with B bins per feature has its full per-node split-gain landscape determined entirely by the $F \times B$ grid of per-bin (g, h) aggregates. No other information from the training set is required to evaluate (1) at every candidate (feature, threshold) pair.*

We define the *split-gain landscape* $\Gamma_D \in \mathbb{R}^{F \times (B-1)}$ as the gain $\mathcal{G}$ that each (feature, bin-boundary threshold) candidate would achieve on dataset D under a fixed prediction state. Lemma 1 is a standard consequence of the histogram GBDT algorithm; we state it here to ground the diagnostic. Its importance is histogram sufficiency: once the prediction state fixes g_i, h_i , Γ_D can be computed in one histogram pass.

B. Split-Landscape Discrepancy

Let $D' = \{(x'_j, y'_j, w'_j)\}_{j=1}^m$ be a weighted selected or synthetic surrogate of D , with $m \ll N$. Synthetic points are binned using the same feature preprocessing and bin boundaries as D ; their labels may be hard labels or soft labels converted to the objective's expected gradients. We quantify the structural gap between D and D' via their landscape distance.

Definition 1 (Split-Landscape Discrepancy). *The relative Split-Landscape Discrepancy of order p between dataset D and surrogate D' is*

$$\text{SLD}_p(D, D') = \frac{\|\Gamma_D - \Gamma_{D'}\|_p}{\max(\|\Gamma_D\|_p, \epsilon)}, \quad (2)$$

using the same bins, objective, regularisation, base score, and feature preprocessing for D and D' . We use $p = \infty$ (worst-case relative split distortion) and the rank-based split-regret variant. Root-SLD compares the root landscape. Reference-tree early-node SLD averages over early nodes of a fixed full-data reference tree, using the reference node memberships. Full-path SLD extends the same comparison along deeper reference-tree paths.

Both variants are computed from histogram statistics under a fixed prediction state, and crucially *no model is retrained on D'* . Figure 1 illustrates how SLD fits into the condensation evaluation pipeline as a no-candidate-retraining alternative to the retrain-and-evaluate loop.

C. Diagnostic Sanity Checks

The diagnostic rests on two standard facts about histogram GBDTs and set coverage. We use them only to motivate what SLD measures; the contribution of the paper is the empirical measurement study across 17 condensation methods.

a) *Margin preservation.*: Let Δ_v be the gain margin between the best-ranked and second-best-ranked split at node v on D . If

$$\text{SLD}_\infty(D, D') < \Delta_v/2, \quad (3)$$

then the argmax split selected by D' coincides with the argmax selected by D at node v , assuming a unique argmax and deterministic tie-breaking. In the measured margin-qualifying cells reported in Section V-A, root-level split agreement equals 1.0. The dose-response experiment (Section V-C) provides the empirical validation.

b) *Coverage guarantee.*: The split-coverage objective is the unweighted fraction of full-data high-gain split thresholds whose adjacent bins are represented by at least one selected or weighted surrogate point. It is binary threshold coverage, not gain-weighted path fidelity. As a standard set-coverage objective, it is monotone submodular under a cardinality constraint, so greedy maximisation has the usual $(1 - 1/e)$ approximation guarantee [29]. This defines the HistDistill-Greedy selector: it

Algorithm 1 Compute Split-Landscape Discrepancy without candidate retraining

Require: Full dataset D , weighted surrogate D' , bins, norm order p , fixed prediction state π

Ensure: $\text{SLD}_p(D, D')$ and split-regret score

- 1: Compute first- and second-order loss statistics for D under π $\triangleright$ root base score or reference-boosters predictions
- 2: Build histogram $H_D \in \mathbb{R}^{F \times B \times 2}$: for each feature f and bin b , store $(\sum_{i: x_{if} \in b} g_i, \sum_{i: x_{if} \in b} h_i)$
- 3: Compute landscape $\Gamma_D \in \mathbb{R}^{F \times (B-1)}$ via (1) applied to every candidate split
- 4: Repeat Steps 1–3 for D' to obtain $\Gamma_{D'} \triangleright$ selected rows reuse weights; synthetic rows are evaluated under π
- 5: $\text{SLD}_p \leftarrow \|\Gamma_D - \Gamma_{D'}\|_p / \max(\|\Gamma_D\|_p, \epsilon)$
- 6: **Split-regret:** let $(f^*, b^*) = \arg \max_{f,b} [\Gamma_D]_{f,b}$ and $(\hat{f}, \hat{b}) = \arg \max_{f,b} [\Gamma_{D'}]_{f,b}$;
- 7: split-regret $\leftarrow \max\left(0, \frac{[\Gamma_D]_{f^*, b^*} - [\Gamma_D]_{\hat{f}, \hat{b}}}{\max([\Gamma_D]_{f^*, b^*}, \epsilon)}\right)$
- 8: **return** SLD_p , split-regret

achieves near-optimal coverage of the split landscape, yet performs poorly by downstream predictive performance (Section V-B). The gap between coverage and predictive fidelity is the central empirical finding of Section V-B.

D. Computing SLD: Algorithm

Algorithm 1 summarises the SLD computation. The key property is that, after the fixed prediction state is available, scoring a candidate requires only one histogram pass and one arithmetic comparison of landscapes; no candidate GBDT is trained. The pass costs $O(NFB)$ to build full-data histograms and $O(mFB)$ for the surrogate, with shared bins and $O(FB)$ landscape memory per node.

IV. EXPERIMENTAL SETUP

A. Methods Evaluated

We evaluate 17 condensation methods spanning five groups; Table I gives the full taxonomy. **Geometry/classical coresets** include Random sampling, Herding [3], and K-center [18]. **GBDT-native subsampling** methods include GOSS [9] and MVS [21], adapted to produce static condensed sets. **Gradient- and score-based selection** methods include GraNd, EL2N [20], CRAIG-style [4], GradMatch-style [19], and gradient sampling (loss-weighted). **Distribution-matching distillation** is represented by DistributionMatching [10] adapted to tabular features. **Gain/landscape-based methods** form our primary family: HistDistill-Greedy (the $(1 - 1/e)$ -optimal coverage selector), HistDistill-Refined and HistDistill-Density (the practical variants optimising landscape fidelity directly), gain sketch, gain-path sketch, and the legacy gain-path refined method. The last three are prior gain-path methods against which we benchmark the refined family.

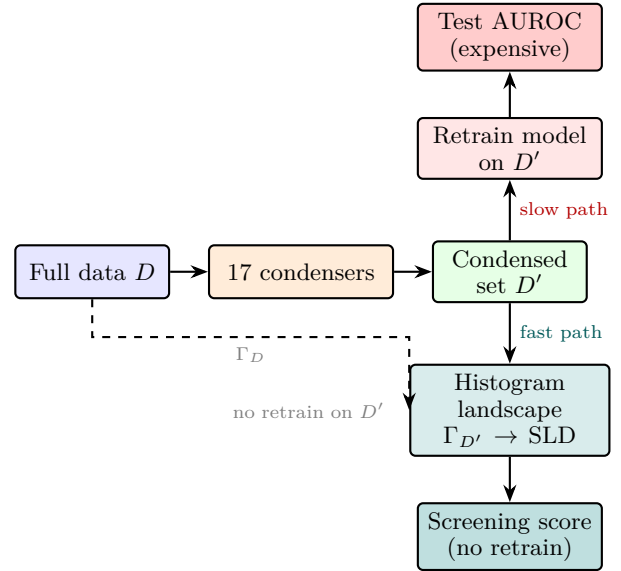


Fig. 1: **SLD evaluation pipeline.** After selecting a condensed set D' from full data D via any of 17 condensers, the fast path computes the candidate’s histogram split-gain landscape $\Gamma_{D'}$ under the same fixed prediction state as Γ_D . It provides a cheap screening signal before retraining on each candidate, in contrast to the slow path of retrain-and-test evaluation.

B. Datasets and Protocol

Our experiments use tabular datasets drawn from OpenML [30], PMLB [31], and UCI-sourced public datasets; the full inventory appears in Table X. The artifact repository lists the exact OpenML/PMLB identifiers and processed split checksums. The core study is on binary classification. For each dataset we condense the training set to five budgets, $\{10, 25, 50, 100, 200\}$ rows, and repeat every (dataset, method, budget) configuration over five random seeds. Across the 26 binary datasets and the five core methods this gives 3,250 retrained-model evaluations for headline performance (Table II) and 650 landscape-fidelity measurements, one per dataset, method, and budget.

The broader 17-method landscape audit uses a lighter but complete diagnostic grid: 26 datasets, two audited budgets (25 and 50 rows), and three seeds, for 156 dataset–budget–seed cells per method. We use this grid for Figure 2 and the method-level ordering analyses. The decomposition study in Table V is a diagnostic subset used only to illustrate the two reference-chain failure modes, so its absolute AUROCs should not be compared directly to the pooled headline values.

Three further studies probe the diagnostic beyond this grid. The robustness study injects calibrated split-gain noise at eight increasing dose levels on all 26 binary datasets, with three seeds and three repeats per level, for 1,872 measurements in total. The generality studies

add five multiclass datasets (scored by one-vs-rest macro-AUROC), five regression datasets (scored by R^2), and a twelve-dataset hyperparameter-transfer benchmark; we report their outcomes in Section VI. A separate scale audit uses Covertypes, SUSY, and HIGGS at training sizes from 20,000 to 10.5M rows with 400-row condensed sets. This audit tests runtime scaling and whether the split-regret ordering survives at deployment scale; it is not used to tune the main 26-dataset benchmark.

C. Metrics

Downstream predictive performance. We score binary tasks by AUROC, multiclass tasks by one-vs-rest macro-AUROC, and regression tasks by R^2 . Every score is computed on a held-out test set with the model retrained on the condensed set.

Fidelity. SLD_∞ is the relative ℓ_∞ landscape distance of (2), and split-regret is the normalised gain lost when a node follows the surrogate’s argmax split rather than the full-data one. Raw SLD is retained in the released tables, but relative SLD and split-regret are the primary metrics because raw landscape scale is dataset-dependent. Both are read off the histogram statistics before retraining on any candidate surrogate.

Error decomposition. To separate the two ways a condensed set can hurt a tree, we evaluate three models in a reference chain. The *structure reference* uses the full-data tree structure with leaves refit under the decomposition protocol. The *leaf-refit model* uses the condensed-set tree structure but refits its leaves on full data, isolating the cost of the structure alone. The *deployed model* is trained only from the condensed set. We define structure error as reference AUROC minus leaf-refit AUROC, and leaf-estimate gap as leaf-refit AUROC minus deployed AUROC. These two gaps add along the reference-to-deployed chain; they are not intended to sum to the ordinary full-data AUROC gap. We also track how representatively the condensed set fills the model’s leaves, defined as the ℓ_1 distance between the condensed- and full-data distributions of training instances across the leaves; we call this the leaf-population mismatch.

D. Learner and Hardware

The reference landscape is built with XGBoost hist trees: 80 boosting rounds, max depth 3, learning rate 0.08, $\lambda = 1$, $\gamma = 0$, subsample 0.9, column subsample 0.9, binary-logistic objective, and fixed seeds. Downstream retraining uses the same XGBoost setting for the GBDT-mechanism results; auxiliary learner slots use LightGBM, CatBoost/fallback histogram boosting, random forests, and MLPs only for robustness checks. For LightGBM we report the current setting explicitly: 120 trees, max depth 4, learning rate 0.06, 15 leaves, subsample and column subsample 0.9, and the library default minimum child samples of 20. This default constrains splits at 10–25-row

TABLE I: The 17 condensation methods. †: adapted from differentiable settings.

Family	Method	Signal / citation
Geometry	Random Herding	Uniform Moments [3]
	K-center	Coverage [18]
GBDT-native	GOSS†	Grad-norm [9]
	MVS†	Min-var [21]
Gradient	GraNd†	Norm at init [20]
	EL2N†	Error L2 [20]
	CRAIG†	Submod. [4]
	GradMatch†	Grad match [19]
	Grad. samp.†	Loss-weighted
Distrib.	DistributionMatching	Distr. match [10]
Gain / landscape	HD-Greedy	$(1 - 1/e)$ coverage
	HD-Refined	Landscape fidelity
	HD-Density	Land.+density
	Gain sketch	Gain sketch
	Gain-path sk. Legacy gp	Gain-path sketch Gain-path refined

budgets, so no conclusion relies on LightGBM-only tiny-budget superiority; the artifact repository includes the budget-aware sweep command for `min_child_samples` $\in \{1, 2, 5, 10, 20\}$. Hardware for the reported local runs is a CPU workstation with no GPU requirement.

E. Reproducibility

All datasets are public, and all preprocessing is deterministic: categorical features are ordinal/one-hot encoded as specified in the scripts, numerical features are binned with shared full-data bin edges, and train/validation/test splits use stratified seeds where labels permit. The single-blind artifact repository, <https://github.com/anonymous-icdm2026/sld-gbdt-condensation>, contains the exact configs, seeds, package versions (Python 3.11.9, NumPy 2.4.4, pandas 2.2.3, scikit-learn 1.8.0, XGBoost 3.2.0, LightGBM 4.6.0, SciPy 1.17.1), processed metric CSVs, plotting scripts, and commands such as `python scripts/19_run_icdm_core.py`, `python scripts/32_run_phase2_experiments.py`, and `python scripts/34_run_large_scale_experiment.py`. The repository is intended to reproduce the reported tables exactly when cached processed data are used, and to closely replicate them from raw public data.

V. RESULTS

A. Does the Landscape Predict Performance?

Finding 1: Method-level split-regret ranks method families in the small-to-mid-size benchmark. Figure 2 and Table III give the method-level audit. Split-regret, computed before candidate retraining, is strongly monotone with eventual AUROC at the method-mean level; gain-path methods sit in the low-regret/high-AUROC corner, while the coverage-optimal HD-Greedy method sits at the opposite extreme. Gradient sampling is the main exception, and the decomposition below shows why: its problem is leaf-value estimation, not split-structure preservation. This 17-method diagnostic

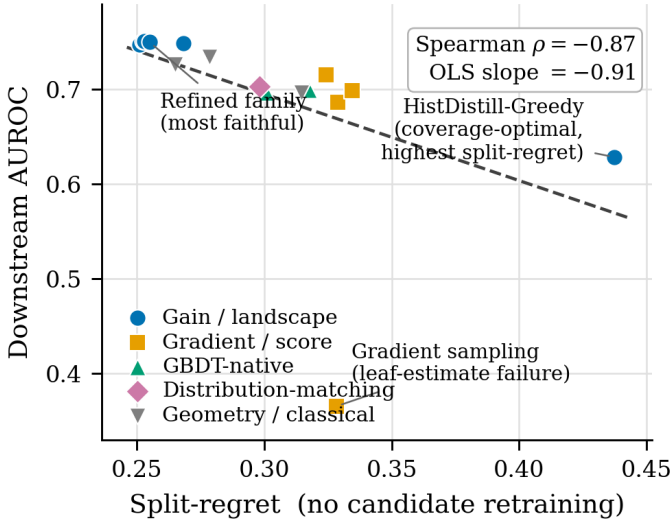


Fig. 2: **No-candidate-retraining method audit.** Each point is one condensation method. Lower split-regret, computed before candidate retraining, tracks higher downstream AUROC over 156 dataset–budget–seed cells (Spearman $\rho = -0.87$, OLS slope -0.91).

TABLE II: Headline performance and fidelity for five core methods (3,250 cells). AUROC mean $\pm$ std; avg. rank lower is better. Speedup is relative to full-data retraining on the same dataset/learner; full-data AUROC is 0.812 ± 0.175 .

Method	AUROC	Rank	SLD $_{\infty}$	Speedup
Full data	0.812 ± 0.175	—	—	1.0 $\times$
Legacy gain-path	0.743 ± 0.186	2.60	18.4	3.9 $\times$
HistDistill-Refined	0.742 ± 0.186	2.68	16.7	3.5 $\times$
HistDistill-Density	0.742 ± 0.186	2.65	17.0	3.5 $\times$
Herdling	0.731 ± 0.176	3.44	34.6	3.8 $\times$
Random	0.724 ± 0.184	3.63	27.7	4.0 $\times$

TABLE III: Seventeen-method diagnostic audit. Values are method means over 156 dataset–budget–seed cells. Lower split-regret and higher AUROC are better.

Method	AUROC	Regret
Gain-path sketch	0.748	0.251
Legacy gain-path	0.751	0.252
HD-Density	0.751	0.253
HD-Refined	0.750	0.255
Random	0.726	0.265
Gain sketch	0.749	0.268
Herdling	0.734	0.279
DM	0.703	0.298
MVS	0.697	0.301
K-center	0.697	0.315
GOSS	0.700	0.317
GradMatch	0.715	0.324
Grad. samp.	0.366	0.328
EL2N	0.686	0.329
GraNd	0.686	0.329
CRAIG	0.699	0.335
HD-Greedy	0.629	0.437

grid uses 26 datasets, two audited budgets, and three seeds, and is separate from the five-budget, five-seed core grid in Table II. The large-scale audit in Section V-E bounds the claim: once high-gain margins are cleared on very large datasets, root split-regret loses rank resolution.

Finding 2: Within a dataset, lower SLD predicts higher AUROC, but pooling across datasets

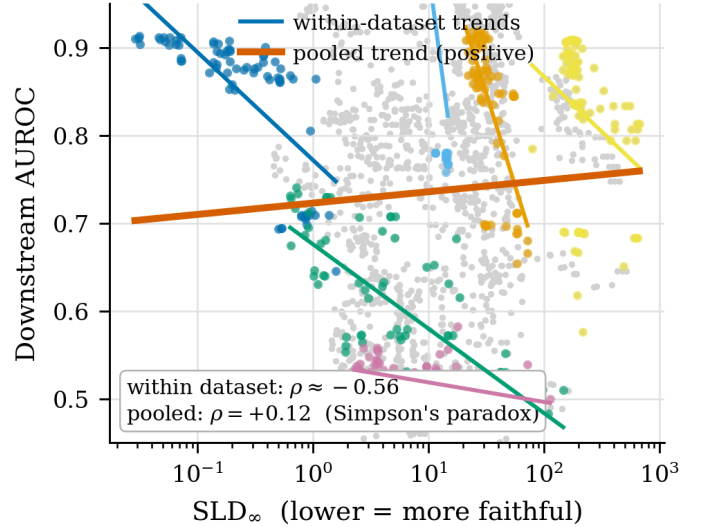


Fig. 3: **Within-dataset SLD trend.** Coloured lines are within-dataset fits on log-scale SLD; the average within-dataset trend is negative. The pooled fit reverses because dataset scale confounds absolute SLD and AUROC.

TABLE IV: Coverage metrics versus AUROC on the two-budget coverage audit. Cov. is high-gain split coverage; Proxy is coverage divided by the proxy upper bound.

Method	Cov25	Proxy25	Cov50	Proxy50	AUROC
HD-Greedy	0.371	1.000	0.539	1.000	0.629
Random	0.082	0.246	0.159	0.299	0.726
HD-Refined	0.075	0.223	0.152	0.286	0.750
HD-Density	0.075	0.223	0.152	0.286	0.751
Herdling	0.059	0.183	0.118	0.229	0.734

is confounded. Figure 3 shows the pattern directly. Within each dataset, lower SLD $_{\infty}$ corresponds to better retrained AUROC; after pooling datasets, the trend can invert because dataset scale affects both SLD magnitude and absolute AUROC. SLD should therefore be used for within-dataset, within-budget method comparison, not as a cross-dataset absolute score.

B. Coverage Is Not Predictive Fidelity

Finding 3: A near-optimal coverage selector is weak by downstream predictive performance. HD-Greedy optimises split-threshold coverage and achieves the expected coverage win, but Figure 4 and Table IV show that this objective is not the one that preserves downstream AUROC. The method lands in the bottom tier of the 17-method audit (Table III); unweighted threshold coverage and gain-weighted landscape fidelity are different objectives.

Finding 4: The coverage failure is structural; gradient sampling fails for a different reason. The reference chain in Figure 5 and Table V separates structure error from leaf-estimate gap. HD-Greedy’s loss comes from choosing the wrong gain-weighted structure, not from poor leaf values after its structure is fixed. Gradient sampling

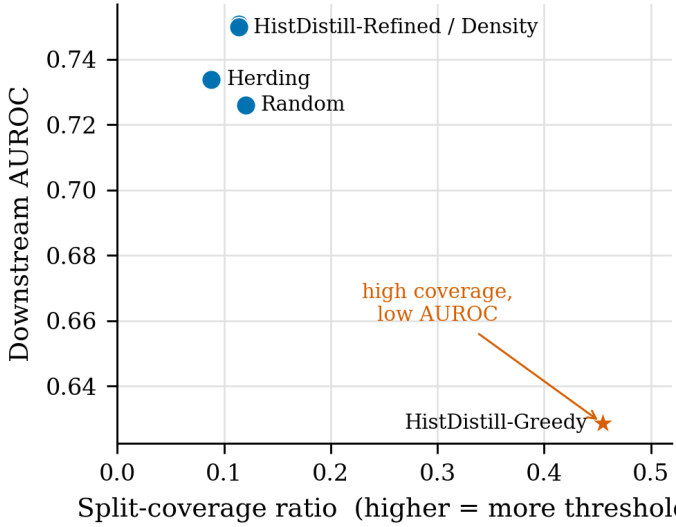


Fig. 4: **Coverage is not predictive fidelity.** HD-Greedy covers the most high-gain split thresholds, but the accurate methods occupy a different region of the coverage–AUROC plane.

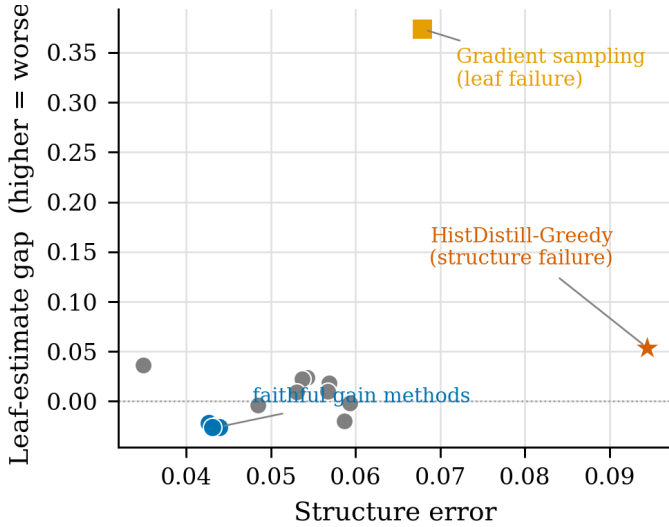


Fig. 5: **Two failure modes.** HD-Greedy is structure-dominated; gradient sampling is leaf-estimate dominated. Faithful gain methods occupy the low-error corner.

is the opposite diagnostic case: its population mismatch is low, but its leaf-estimate gap is high, pointing to distorted within-leaf gradient/Hessian or reweighting statistics.

C. How Much Can Structure Be Perturbed?

Finding 5: GBDT structure shows a margin-linked robustness pattern. Figure 6 and Table VI inject calibrated split-gain noise into the full-data landscape. AUROC and split agreement stay flat while the margin condition in (3) holds; beyond that point, agreement drops first and AUROC degrades mildly. The result explains why aggressive condensation can work when the dominant

TABLE V: Error decomposition on the diagnostic subset used to illustrate both failure modes. Absolute AUROCs differ from the pooled values in Table II and Figure 2. Structure error is reference minus leaf-refit; leaf gap is leaf-refit minus deployed.

Method	Dep.	Leaf	Str. Err	Leaf Gap	Pop.
HD-Greedy	0.643	0.696	0.094	0.054	0.243
Grad. samp.	0.349	0.723	0.068	0.374	0.133
DM	0.720	0.756	0.035	0.036	0.235
K-center	0.713	0.737	0.054	0.024	0.196
MVS	0.724	0.734	0.057	0.010	0.088
GradMatch	0.734	0.732	0.059	−0.002	0.113
HD-Refined	0.773	0.747	0.044	−0.026	0.060
Legacy gp	0.774	0.748	0.043	−0.026	0.068

Full-data AUROC = 0.826; reference AUROC = 0.791.

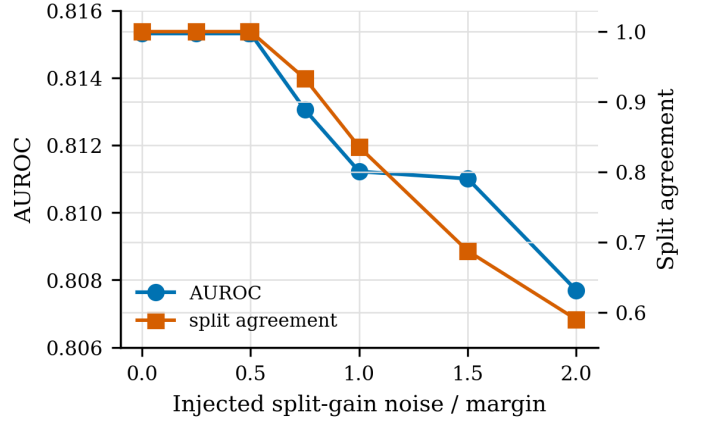


Fig. 6: **Robustness pattern.** Split agreement drops after dose 0.5; AUROC degrades mildly as calibrated split-gain noise exceeds the margin.

TABLE VI: Dose-response: 26 binary datasets, 1,872 cells. Eight injected-noise levels are grouped into five displayed rows.

Dose	AUROC	Δ AUC	Agree.	Regret
≤ 0.50	0.815	0.000	1.000	0.000
0.75	0.813	−0.002	0.933	0.011
1.00	0.811	−0.004	0.836	0.030
1.50	0.811	−0.004	0.687	0.069
2.00	0.808	−0.008	0.590	0.101

split margins are preserved, and why corrupting high-gain structure is disproportionately costly.

D. Performance, Fidelity, and Cost

Finding 6: HistDistill-Refined and -Density are practically tied with the strongest prior selector while being the most faithful. Tables II and IX carry the performance, fidelity, and cluster-aware paired comparison numbers. The interpretation is practical parity at the top: the refined family matches the legacy gain-path selector in AUROC while achieving the lowest landscape discrepancy among the core methods. The Refined-vs.-Legacy mean difference is operationally negligible and its dataset-clustered confidence interval crosses zero. At the

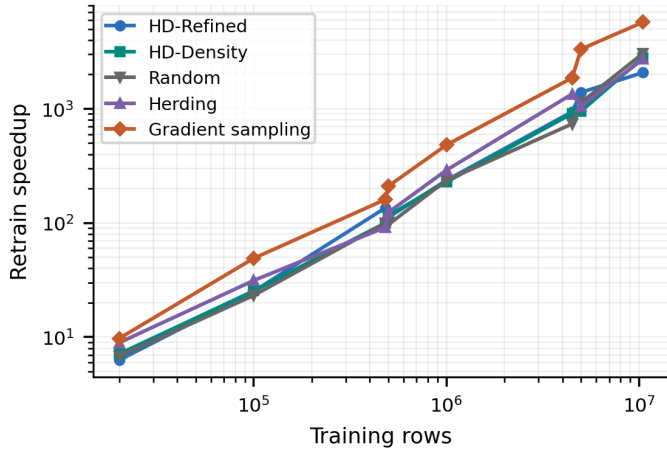


Fig. 7: **Large-scale retraining payoff.** Log-log speedup for 400-row condensed retraining on Covertypes, SUSY, and HIGGS. Lines connect median speedup at each observed training size.

studied small-data budgets, condensed retraining gives a median 3–4 $\times$ speedup, with median condensed training under half a second in the recorded runs; SLD itself costs one histogram pass.

E. Does the Diagnostic Hold at Scale?

Scale audit: the cost payoff grows with N , while ordering is dataset-dependent. Figure 7 and Tables VII and VIII report a real large-data audit using 400-row condensed sets. Retrain speedup rises from single digits at 20k rows to hundreds or thousands at million-row scale. Random has the best large- N AUROC point estimate on all three datasets, but the three-dataset paired audit is not powered to claim statistical superiority: Random’s mean advantage over HD-Density is 0.034 AUROC and over HD-Refined is 0.035 AUROC, with two-sided Wilcoxon $p = 0.25$ in both comparisons. The practical reading is convergence among reasonable condensers, not a new winner-take-all method ranking.

The HIGGS case is informative rather than contradictory. Several methods have zero root split-regret there, yet their AUROCs differ. This is exactly the margin story from Figures 3 and 6: at very large N , the dominant root split is so well separated that many surrogates clear the same margin, and leaf-estimate quality or deeper structure becomes the discriminating factor. Root-level regret remains useful for rejecting severe failures, but not for ranking reasonable surrogates once the top split is saturated. A rule that only rejects the obvious gradient-sampling failure retains the winner in all scale cells while saving less than 2% of already-small condensed-search time; the real cost win comes from training on hundreds of rows rather than millions.

TABLE VII: Large-scale audit at the largest training size per dataset. ρ is Spearman correlation over the five audited methods; negative ρ means lower split-regret aligns with higher AUROC.

Dataset	Rows	ρ	Top speedup
Covertypes	0.48M	−0.36	160 $\times$
SUSY	4.50M	−0.97	1,864 $\times$
HIGGS	10.50M	0.11	5,734 $\times$

TABLE VIII: Large-scale AUROC at the largest training size. Full-data AUROCs are 0.897, 0.873, and 0.803 for Covertypes, SUSY, and HIGGS. Random wins the condensed point estimates, but the three-dataset paired audit is underpowered ($p = 0.25$ against HD-Refined and HD-Density).

Method	Cov.	SUSY	HIGGS
Full data	0.897	0.873	0.803
Random	0.792	0.839	0.687
HD-Refined	0.747	0.818	0.648
HD-Density	0.781	0.794	0.641
Herding	0.719	0.780	0.450
Grad. samp.	0.377	0.219	0.364

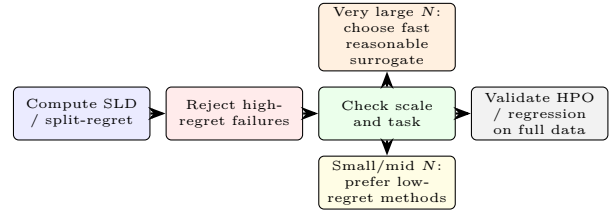


Fig. 8: **Operational workflow implied by the measurements.** SLD is most useful as a cheap screening diagnostic: reject high-regret failures, prefer low-regret methods only when margins are not saturated, and validate HPO, regression, and final deployment choices on full data.

TABLE IX: Dataset-clustered core-method comparisons. Δ is the paired mean AUROC difference over 3,250 cells; confidence intervals bootstrap the 26 dataset clusters.

Method	Comparator	Δ	95% CI	DS wins
HD-Refined	Legacy gp	−0.00056	[−0.0019, 0.0008]	11/26
HD-Density	Legacy gp	−0.00056	[−0.0016, 0.0005]	9/26
HD-Refined	Herding	0.01183	[0.0037, 0.0204]	18/26
HD-Refined	Random	0.01807	[0.0124, 0.0233]	24/26

VI. WHERE THE LENS STOPS: SCOPE AND LIMITATIONS

The SLD diagnostic is designed around the inductive bias of histogram GBDTs. Three scopes delimit where that design leads to weak or negative results.

Regression is structure-dominated, and no condenser fixes it. On the five regression datasets, the structure reference—the full-data tree structure with leaves refit under the decomposition protocol—achieves $R^2 \approx 0.63$. The leaf-refit model, which uses each condensed set’s

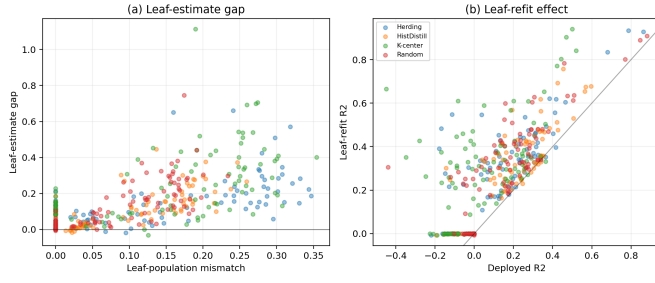


Fig. 9: **Regression failure modes.** Left: leaf-population mismatch tracks leaf-estimate loss. Right: refitting leaves on full data helps, but the remaining structure gap dominates.

structure but refits leaves on full data, falls to $R^2 = 0.28$ – 0.29 . The deployed condensed models then reach only $R^2 \approx 0.16$ for HistDistill and 0.05 for K-center. Thus, for HistDistill, the total gap from the structure reference to the deployed model is about 0.47 ; of that gap, about 0.35 is structural ($0.63 - 0.28$) and about 0.12 is due to leaf-value estimation ($0.28 - 0.16$). No method we evaluate closes the structure gap: the best deployed condensed R^2 is 0.16 , well below even the leaf-refit baseline of 0.28 . Figure 9 shows this pattern. The SLD diagnostic correctly *diagnoses* this failure, since the condensed sets have high SLD_∞ relative to full-data regression baselines, but it does not cure it. Regression tasks appear to require structure fidelity that current condensers cannot achieve at the budgets we study.

HPO transfer is weak. We evaluate whether condensed sets preserve hyperparameter rankings: if a configuration performs well on the condensed set, does it also perform well on the full data? Across 12 binary datasets and a standard HPO search grid, the top-3 Jaccard between condensed- and full-data HPO results is 0.042 , top-1 match rate is 0.083 , and the Spearman rank correlation of configuration performance across the two settings is 0.145 . The median speedup from running HPO on the condensed set is $1.57\times$: modest, and not enough to compensate for the near-zero rank preservation. We also report a mean test-AUROC gap of 0.00125 between the best configuration found on the condensed set and the best on the full data, but this small gap is misleading given the near-zero Jaccard: the condensed set finds a *different* configuration that happens to be competitive, not the same one. Condensed HPO search should therefore be followed by a full-data check of the top configurations before deployment.

No-retraining selection of the single best method remains weak. SLD ranks method families well at the aggregate level (Spearman -0.87), but selecting the *single best* method for a new dataset based only on SLD is harder. The top-3 success rate of the SLD-based selector is 0.436 across 156 dataset-budget-seed cells, and the top-1 rate is 0.122 . The split-regret selector performs simi-

larly. The landscape score is useful for screening out bad condensers and ranking families, but the gap between the top-2 or top-3 methods is often smaller than the within-dataset variance of SLD, making single-method selection unreliable. Future work could combine the landscape score with dataset-level meta-features to improve per-instance selection.

Multiclass classification is a thin positive extension (HistDistill 0.910 macro-AUROC vs. Random 0.897), but taken together these scopes mark the boundary of the diagnostic: SLD is built around the GBDT split-gain objective, and its predictive value erodes as a task drifts away from that inductive bias.

VII. CONCLUSION

Measuring mechanism, not just downstream predictive performance, reveals things that a retrain-and-compare protocol cannot. For gradient-boosted decision trees on tabular data, the split-gain landscape is a central mechanism for tree-structure selection: it determines which features and thresholds the booster commits to before a leaf value is estimated. Comparing landscapes between a full dataset and its surrogate, through the Split-Landscape Discrepancy, yields a no-candidate-retraining diagnostic whose method-level split-regret correlates with downstream AUROC in the small-to-mid-size benchmark at Spearman -0.87 . The scale audit qualifies this claim: once very large datasets make the dominant split margins easy to preserve, random sampling becomes difficult to beat and root split-regret becomes a screening signal rather than a fine-grained ranker.

The most practically consequential finding is that *coverage is not predictive fidelity*. A condensation method with a provable near-optimal guarantee on unweighted split-threshold coverage is simultaneously among the weakest downstream performers in the pooled audit. The reason is structural: unweighted coverage fills the landscape broadly, while AUROC depends on gain-weighted, Hessian-weighted, path-sensitive structure preservation. The reference-chain decomposition makes this precise: the coverage-optimal method’s failure is a structure failure rather than a leaf-value failure.

We also delimit the scope of these findings. Regression tasks are structure-dominated in a way that current condensers cannot overcome. Hyperparameter transfer from condensed sets to full-data rankings is weak. At deployment scale, reasonable condensers can converge in root split-regret while still differing in leaf estimates and deeper structure. These boundaries reinforce the role of SLD as a measurement instrument rather than just a performance metric.

The applied workflow is therefore direct: before choosing a condensation strategy, compute SLD in a histogram pass and use split-regret to reject bad candidates. This screens out poor surrogates before retraining on every candidate; when N is very large and several candidates clear the

TABLE X: Dataset inventory (binary; public via OpenML/PMLB). *: focused binary suite.

Dataset	N	F	Dataset	N	F
adult*	20k	14	spambase*	4,601	57
australian*	690	14	agaricus	8,145	22
bank_mktg*	20k	16	credit_appr.	690	15
breast_w*	699	9	electricity	20k	8
credit_g*	1k	20	gametes (6)	1,600	20
diabetes*	768	8	hill_val.(2)	1,212	100
pc1*	1,109	21	kr_vs_kp	3,196	36
phoneme	5,404	5	pc3, pc4	~1,500	37
qsar_biodeg	1,055	41	sick	3,772	29
smoke	1,500	12	tic_tac	958	9

dominant split margins, choose among the reasonable surrogates by budget, speed, and downstream validation. Coverage metrics should not be used as an AUROC proxy, and for regression tasks or HPO transfer condensed sets should be treated as approximations to validate rather than substitutes for the full data. We release the diagnostic and benchmark in the single-blind artifact repository to make this workflow reproducible and extensible to new condensation methods as they emerge.

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